USED CAR MARKET ANALYSIS

Project Report

Tools: Microsoft Excel and Tableau

An analysis of car listings, prices, mileage, manufacturers and vehicle characteristics.

1. Project Overview

This project analyses a used-car listings dataset to understand the characteristics of the listed vehicles and how their prices differ across manufacturers, mileage, fuel types, transmission types, usage status and locations. I used Microsoft Excel to inspect and clean the dataset, then imported the cleaned data into Tableau to create KPIs, charts, filters and three interactive dashboards.

During the analysis, I asked business questions about the dataset and used the available fields to answer them through visualisations. The main objectives were to understand the overall listed car value, compare average prices, examine mileage and price, and identify the manufacturers with the highest listing counts and total listed values.

2. Dataset and Data Cleaning

The dataset contained 10 columns: title, odometer, location, isimported, engine, transmission, fuel, paint, price and year. I first reviewed the dataset in Excel to check for duplicate records. I found one exact duplicate for a Lexus RX 350; the values in every column were identical to the other record. I removed that duplicate and then imported the cleaned dataset into Tableau.

After importing the data, I reviewed the fields and prepared the measures and dimensions needed for the analysis. I created the KPIs and used different chart types to answer the questions I had about the car listings.

3. Key Performance Indicators

The four main KPIs used in the project were:

- Total Car Value – the combined value of all listed car prices.
- Average Price Listed – the average asking price of the cars in the dataset.
- Average Odometer – the average mileage recorded for the listed cars.
- Average Car Age – the average age of the listed cars, calculated from their manufacturing year.

The total car value represents the sum of advertised prices in the dataset. It is not confirmed sales revenue because the dataset contains listings rather than completed transaction records.

4. Dashboard 1 – Car Market Overview

The first dashboard gives an overview of the car market. It includes the main KPIs, average price by manufacturer, usage status by number of cars, car listings by location and average price by engine type.

Average Price by Manufacturer

Rolls-Royce had the highest average listed price, followed by Mercedes-Benz/52, Land Rover and Bentley. Fiat had the lowest average listed price among the manufacturers shown.

Usage Status by Number of Cars

I compared the number of foreign-used, locally-used and new cars. Foreign-used cars had the highest number of listings, followed by locally-used cars and then new cars.

Car Listings by Location

Lagos had the highest number of car listings, followed by Abuja and Ogun State. This shows that the listings in the dataset were concentrated mainly in Lagos.

Average Price by Engine Type

The 12-cylinder engine category had the highest average listed price, followed by the 8-cylinder category. The 2-cylinder category had the lowest average listed price.

5. Dashboard 2 – Car Mileage and Fuel Analysis

The second dashboard focuses on transmission distribution, average price by fuel type, usage status by average price, and the relationship between odometer readings and listed car prices. I also added manufacturer and car-model filters to compare vehicles from the same manufacturer and model.

Transmission Distribution

Automatic cars accounted for 1,378 listings, representing approximately 95% of the dataset. Manual cars accounted for 72 listings, representing approximately 5%. This shows that automatic transmission cars made up the majority of the listings.

Average Price by Fuel Type

Petrol cars had a higher average listed price than diesel cars in the dataset.

Usage Status by Average Car Price

New cars had the highest average listed price, followed by foreign-used cars. Locally-used cars had the lowest average listed price.

Does Odometer Affect Car Price?

To answer this question, I used a scatter plot to compare odometer readings with listed prices. I added Car Manufacturer and Car Model filters so that I could compare vehicles from the same manufacturer and model. The overall pattern showed an inverse relationship between mileage and price: as the odometer reading increases, the listed price generally tends to decrease.

However, the Lexus LX 570 comparison showed an exception. The 2014 Lexus LX 570 had a higher odometer reading and was listed at approximately NGN 110 million, while the 2018 model had a lower odometer reading but was listed at approximately NGN 58 million. This comparison shows that mileage is not the only factor affecting a car's listed price. The model year and other characteristics can also influence the price.

6. Dashboard 3 – Car Manufacturer Analysis

The third dashboard compares manufacturers by average mileage, number of car listings and total listed car value. It includes rankings for the top and bottom manufacturers across these measures.

Manufacturers by Average Mileage

Dodge had the highest average mileage, followed by Audi. Toyota did not appear among the top 10 manufacturers by average mileage. This means Toyota's strong performance in listing count and total listed value did not come from having the highest average mileage.

Manufacturers by Number of Listings

Toyota had the highest number of car listings, with 699. Lexus followed with 175 listings, while Mercedes-Benz had 147 listings.

Manufacturers by Total Listed Car Value

Toyota had the highest combined listed car value at approximately NGN 4.33 billion. Lexus followed at approximately NGN 2.69 billion, while Mercedes-Benz had approximately NGN 2.05 billion. These figures represent the combined asking prices of listed vehicles, not confirmed revenue from completed sales.

7. Key Findings

- One exact duplicate Lexus RX 350 record was found and removed during the Excel data-cleaning stage.
- Foreign-used cars made up the largest usage-status group in the dataset.
- Automatic cars represented approximately 95% of listings, while manual cars represented approximately 5%.

- Lagos had the highest number of car listings, followed by Abuja and Ogun State.
- Rolls-Royce had the highest average listed price among the manufacturers shown, while Fiat had the lowest.
- The 12-cylinder engine category had the highest average listed price, while the 2-cylinder category had the lowest.
- Petrol cars had a higher average listed price than diesel cars.
- New cars had the highest average listed price by usage status, followed by foreign-used and locally-used cars.
- The mileage and price comparison generally showed an inverse relationship, although the Lexus LX 570 example showed that this pattern does not apply to every individual listing.
- Dodge had the highest average mileage, while Toyota had the highest number of listings and the highest total listed car value.

8. Recommendations

- Marketplace operators can give particular attention to Lagos because it had the highest number of listings in the dataset, while also maintaining visibility in Abuja and Ogun State.
- Toyota, Lexus and Mercedes-Benz can be prioritised when reviewing listing volume and total advertised value because they had the highest listing counts and combined listed values in the manufacturer comparison.
- When comparing or pricing vehicles, consider mileage together with the model year, manufacturer, model, fuel type, usage status and other vehicle characteristics. The Lexus LX 570 example shows why mileage should not be considered on its own.
- The strong share of automatic listings can guide how vehicle inventory categories and search filters are presented on a car marketplace.
- Use average price comparisons by manufacturer and engine type to understand differences in the advertised market, while considering the number of listings behind each average.

9. Conclusion

The project provided an overview of the used-car listings and showed how prices and listing patterns differ by manufacturer, location, usage status, engine type, fuel type, transmission and mileage. Foreign-used and automatic cars made up the largest shares of the listings, while Lagos had the highest listing count. Toyota led in both the number of listings and total listed car value. The mileage analysis generally showed that cars with higher odometer readings tend to have lower listed prices, although the Lexus LX 570 comparison demonstrated that vehicle year and other characteristics can change this pattern.

Excel was used for the initial data inspection and duplicate removal, while Tableau was used to calculate the KPIs, explore the data and present the findings through three interactive dashboards. All monetary values in this report are expressed in NGN.